Open to internship · July 2026

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SKILLS

MACHINE LEARNING & LLMS

- LLM architecture (GQA · RoPE · SwiGLU) — Proficient
- Pre-training pipeline (data · curriculum · BPE) — Proficient
- Fine-tuning (SFT) — Proficient
- Mixed-precision training (BF16/FP8) — Intermediate
- Inference & deployment (HF) — Intermediate

NUMERICAL & OPTIMIZATION

- Numerical optimization (constrained) — Advanced
- Optimal control · ODE solvers (shooting · MPC) — Proficient

FRAMEWORKS & TOOLS

Python · PyTorch · HuggingFace · Transformer Engine · Flash Attention · FP8 / BF16 · JAX/Flax · NumPy · scikit-learn · Julia · SQL · Git · Linux · HPC

LANGUAGES

French: Native
English: Professional
Spanish: Basic

Nicolas Bailliet

LLM / Deep Learning — model training, architecture & inference optimization

Seeking 5-month internship · From July 2026 · Paris / Remote

PROFESSIONAL PROFILE

Third-year Computer Science student at ENSEEIHT with a strong background in mathematics and optimization. Built and deployed **HilbertLM** — two transformer LLMs (135M & 163M params) trained entirely from scratch on 20B tokens using Flash Attention 3 & FP8 quantization on a single RTX 5080 (~70h). Full pipeline: custom BPE tokenizer, 4-stage curriculum pre-training, SFT fine-tuning, and HuggingFace deployment with live chatbot. Seeking a role in **LLM training, architecture, or inference optimization**.

PROFESSIONAL EXPERIENCE

Research Engineer Intern — Optimal Control & Scientific Computing — Jun-Aug 2025

IRIT / ENSEEIHT · Toulouse

- Enhanced and validated an open-source Julia optimal-control solver library, across two axes: software maintenance (bug fixing) and numerical validation.
- Built an automated benchmark looping over 15 control problems: convergence checks for both the solver and JuMP (reference Julia modeling language), initialization comparison, and relative error in the L^p norm — run at p=2 — on the state (x) and control (u) between the two solvers.
- Validated the Lunar Landing maneuver (minimum-fuel soft touchdown), reproducing known analytical solutions with high precision and confirming the fix of critical bugs.
- Investigated an Earth-to-Moon transfer modeled as the Restricted Three-Body Problem (CR3BP); implemented regularization to smooth singularities and scoped solver limitations for future work.
- Contributed unit tests & API documentation via GitHub (commits, Pull Requests, peer reviews) — positive team feedback for surfacing critical bugs. Stack: Julia · JuMP.

EDUCATION

Computer Science & Engineering — ENSEEIHT (INP Toulouse) — 2023–Present

ML algorithms, numerical methods, optimization, HPC, big data systems.

Preparatory School, Maths & Physics — Lycée Chaptal, Paris — 2020–2023

PROJECTS

★ HilbertLM (135M & 163M)

Live demo ↗ (baillietn.github.io/HilbertLM-Lab/)
HuggingFace ↗ (huggingface.co/collections/baillietn/hilbertlm)

135M & 163M params · 20B tokens trained · ~70h · RTX 5080 · 79K tok/s throughput · 4 models on HF Hub

Python · PyTorch · Transformer Engine · Flash Attention 3 · FP8 · HuggingFace · BPE

Two decoder-only transformers built entirely from scratch. 30-layer architecture: GQA (9Q/3KV heads), RoPE, SwiGLU — custom BPE tokenizer (49K vocab). 4-stage curriculum pre-training (FineWeb-Edu, Python-Edu, FineMath, Cosmopedia-v2) + SFT on 100M conversational tokens (SmolTalk). Loss landscape visualization via PCA on checkpoints. Deployed as Base & Chat variants (BF16 + FP8) on HuggingFace Hub with streaming chatbot interface.

Soprasteria–ENSEEIHT · Water Quality IBD Prediction

Python · TensorFlow/Keras · scikit-learn · pandas

Neural network predicting Biological Diatom Index for river ecosystem monitoring. 70+ environmental features from Naïades DB (2000–2025); BatchNorm & Dropout; anomaly detection & feature importance analysis.

ENM–ENSEEIHT · Deep Learning Practical Modules

Python · PyTorch · Jupyter

Designed 8 progressive ML modules covering: NN fundamentals, CNNs, attention mechanisms, transformers, semantic segmentation, GANs/WGAN, quantile regression, learning-to-rank (RankNet + curriculum learning).

Stochastic Filtering & Data Assimilation

Python · JAX · Flax · NumPy

4 notebooks: Kalman Filter, Ensemble Kalman Filter (Lorenz dynamics), score-based diffusion models (JAX/Flax), PageRank via Markov chains.

Optimal Control — Course Materials & Orbital Transfer

Julia · DifferentialEquations.jl · NLsolve.jl · ForwardDiff.jl

6 practical modules: ODEs, shooting methods, MPC, automatic differentiation, Runge-Kutta schemes. Orbital transfer optimization (Hohmann transfer, 3D trajectory modeling).